AIML-401: Deep Learning — Revision Study Guide

Topic: **Forward Propagation**

Candidate ID: STU-807	**Target Exam:** Midterm IA-2 & SEE Exam
Curriculum Unit: Unit 5: Generative Models & Adversarial Training	**Compiled:** Nov 03, 2026 - 00:05
Engine: Autonomous Python Script Execution	**Platform:** PACD-RAG Academic Intelligence

1. Fundamental Concept & Definition

Forward propagation is a fundamental concept in neural networks, where the input data is passed through layers of neurons to produce an output. It involves the sequential application of linear transformations and activation functions to the input data.

2. Mathematical Formulation & Generalization Theory (No Code)

Forward propagation can be mathematically represented as follows:

$$Y = f(WX + b)$$

Where Y is the output, W is the weight matrix, X is the input vector, b is the bias vector, and f is the activation function.

Mathematical Constraint: *Theoretical & Mathematical Formulation verified: Generalization risk bound $P(|R(h) - R_{emp}(h)| > eps) <= 2 exp(-2 N eps^2)$.*

3. 4-Phase Execution Pipeline

Phase	Name	Mathematical / Code Operation	Key Objective
Input Layer	Input Data	Receive raw input data	Prepare the initial data for processing
Hidden Layers	Linear Transformation	Apply weights and biases to the input data	Transform the input data into a new space
Hidden Layers	Activation Function	Apply an activation function to the transformed data	Introduce non-linearity to the model
Output Layer	Output Calculation	Pass the output through the final layer	Generate the final output of the neural networ

4. High-Yield Question & Pitfall Checklist for Exams

Top Exam Trap: Failing to account for the activation function during forward propagation can lead to incorrect outputs, as it introduces non-linearity into the model.

Expected Question (8 Marks): *'Explain the process of forward propagation in a neural network and describe the role of the activation function in this process.'*

Marking Rubric:

• Understanding of forward propagation process (2 marks)
• Explanation of activation function's role (2 marks)
• Clarity and coherence of the answer (2 marks)